James Streett V

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Personal Website & Interactive Resume: Link

Summary

Full Stack Software Engineer with a B.S. in Machine Learning from UC San Diego and hands-on experience co-designing, scaling, and deploying distributed microservices for multi-tenant B2B platforms. Proven track record architecting high-throughput data ingestion pipelines on AWS, establishing CI/CD workflows with GitHub Actions, DSP Telemetry Processing and optimizing relational databases (PostgreSQL), to process multi-million record datasets.

Skills

Core Engineering: Python, Java, C++, C#, MATLAB, Git, SQL, PostgreSQL, REST APIs, Bash
Cloud & Data Infra: AWS, OpenSearch, Docker, Kubernetes, Github Actions, CI/CD workflow
Machine Learning: PyTorch, Tensorflow, HuggingFace, LLMS, LoRA, Claude, Cursor, Signal Processing, Deep Learning

Experience

Software Engineer, *CreateBase* - Los Angeles, CA (Dec 2025 – Present)

- **Distributed Systems & Multi-Tenant Architecture:** Co-designed and scaled a multi-tenant B2B platform supporting artists, publishers, and admins; leveraged OpenSearch and AWS to orchestrate high-throughput indexing and fuzzy matching across chaotic multi-million record metadata catalogs.
- **CI/CD & Container Orchestration:** Built automated CI/CD pipelines using GitHub Actions and Docker to build, test, and deploy microservices to AWS ECS Fargate, streamlining developer workflows and containerized deployments.
- **High-Throughput Pipelines & DB Optimization:** Architected asynchronous backend ingestion pipelines and optimized relational database schemas (PostgreSQL) to handle heavy concurrent read/write loads from cross-platform auditing, significantly reducing API latency for global users.
- **Data Pipelines & Database Performance:** Built asynchronous ingestion services and tuned SQL query performance (PostgreSQL) to support real-time data auditing under heavy concurrent API traffic.

Skills Applied: Python, OpenSearch, AWS, Kubernetes, PostgreSQL, REST APIs, Docker, Github Actions

Machine Learning Engineer Intern, *VoiceTune AI* - Dublin, Ireland (Jul 2025 – Sep 2025)

- **Generative AI & Fine-Tuning:** Fine-tuned open-source audio models and LLMs using Hugging Face and LoRA, building custom pipelines for voice modulation and generative audio applications.
- **DSP & Feature Extraction:** Built Python and Librosa feature-extraction pipelines, leveraging DSP techniques to process complex acoustic data for downstream machine learning tasks.
- **R&D Collaboration:** Partnered with researchers at Trinity College Dublin to integrate machine learning workflows and telemetry analysis tools into core R&D infrastructure.

Skills Applied: Python, Hugging Face, LoRA, LLMs, Librosa, DSP, Generative Audio

Bioacoustic Data Engineer, *San Diego Navy Base* - San Diego, CA (Dec 2024 – Jun 2025)

- **High-Bandwidth Data Movement:** Analyzed massive, complex hydro-acoustic data streams to identify, catalog, and isolate unique time-series data signatures. DSP Telemetry processing across buoy networks
- **Workflow Automation:** Defined parameters and optimized data classification algorithms for massive sensory libraries, streamlining edge data pipelines for classification workflows.

Skills Applied: Python, MatLAB, DSP, Telemetry, Acoustic Analysis

Projects

DolphinBox Bioacoustic Classification, *Award Winning Research Project* - (Jun 2025)

- Built a real-time embedded system that classifies bioacoustic signals into spectrograms and MFCC feature vectors for dolphin communication classification.
- Trained a multi-label convolutional neural network (CNN) achieving 94% classification accuracy, optimized for live, low-latency target identification on local hardware.
- **Awarded the Michael L. Sherman ICAM Research Award** (Sole recipient across graduating class).

Skills Applied: Python, PyTorch, DSP, Acoustic Engineering

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CNN Music Classification & Song Recommendation System, *Research Project* - (Mar 2025)

- Architected a 3-layer CNN in TensorFlow/Keras to classify audio into 10 music genres using 60 extracted spectral and MFCC features via Librosa, achieving 73% test accuracy.
- Solved data leakage issues present in benchmark Kaggle pipelines by engineering a strict track-level split, ensuring audio integrity across train/test datasets.
- Developed a content-based recommendation engine leveraging high-dimensional feature embeddings extracted from the CNN flatten layer and cosine similarity metrics.

Skills Applied: Python, TensorFlow, Keras, Librosa, Feature Extraction, Deep Learning, Cosine Similarity

Certifications

- AWS Certified Solutions Architect - Associate (Expected Completion: Dec 2026)
- Dante Level 1 & Level 2 Certified: Audinate (Issued: Nov 2025)

Education

University of California San Diego

- *B.S.* in Machine Learning - *(Minor* in Computer Science, GPA: 3.8)
- *B.A.* in Interdisciplinary Computing & the Arts - *(DSP/Audio Engineering)*